

## Histograms

- 1D Histograms
- 3D Histograms
- Equalisation
- Histogram Comparison
- Back Projection

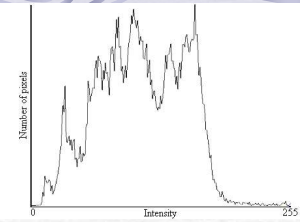
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## 1D Histograms



- Determine the frequency of brightness values

- Initialise:
  - $h(z) = 0$  for all values of  $z$
- Compute:
  - For all pixels  $(i,j)$ :  $h(f(i,j))++$

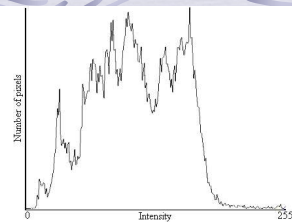
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## 1D Histograms



- Is this useful?

- Global information
- Useful for classification ?
- Not unique

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## 1D Histograms

```
Mat display_image;
MatND gray_histogram;
const int* channel_numbers = { 0 };
float channel_range[] = { 0.0, 255.0 };
const float* channel_ranges = channel_range;
int number_bins = 256;
calcHist( &gray_image, 1, channel_numbers, Mat(),
          gray_histogram, 1, &number_bins, &channel_ranges );
OneDHistogram::Draw1DHistogram( &gray_histogram, 1,
                                display_image );
imshow("Greyscale histogram", display_image);
```

Histograms

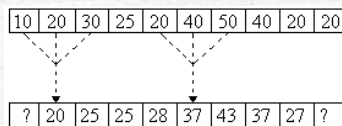
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## 1D Histograms – Smoothing

- Local minima and maxima are useful.
- But how can we deal with noise though?



- Smooth the histogram...

- For all values  $v$ :  $h_{\text{new}}(v) = (h(v-1) + h(v) + h(v+1)) / 3$

- What do we do at the ends?

- Do not compute OR Wraparound OR Duplicate values OR Reflect values OR Constant values ??

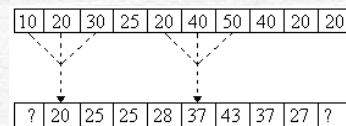
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## 1D Histograms – Smoothing in OpenCV



```
MatND smoothed_histogram = histogram[channel].clone();
for(int i = 1; i < histogram[channel].rows - 1; ++i)
    smoothed_histogram[channel].at<float>(i) =
        (histogram.at<float>(i-1) + histogram.at<float>(i) +
         histogram.at<float>(i+1)) / 3;
```

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## 1D Histograms – Colour Histograms

- Determine histograms for each channel independently...
- Choice of colour space...

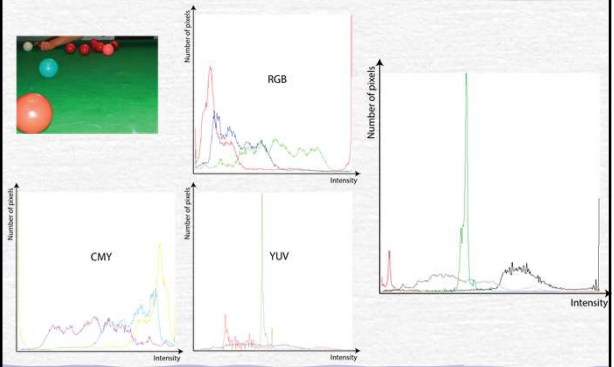
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## 1D Histograms – Colour Histograms



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## Colour (1D) Histograms in OpenCV

```
MatND* histogram = new MatND[image.channels()];
vector<Mat> channels(image.channels());
split( image, channels );
const int* channel_numbers = { 0 };
float channel_range[] = { 0.0, 255.0 };
const float* channel_ranges = channel_range;
int number_bins = 64;
for (int chan=0; chan < image.channels(); chan++)
    calcHist( &(channels[chan]), 1, channel_numbers, Mat(),
             histogram[chan], 1, &number_bins, &channel_ranges );
```

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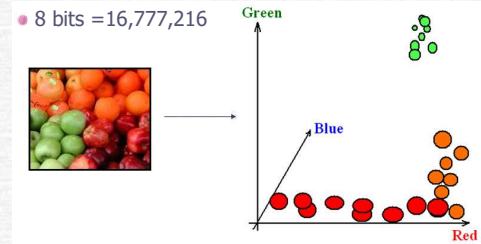
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## 3D Histograms

- Channels are not independent
- Better discrimination comes from considering all channels simultaneously
- Number of cells?
  - 8 bits = 16,777,216



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## 3D Histograms

- Reduce quantisation
  - 6 bits = 262,144
  - 4 bits = 4,096
  - 2 bits = 64



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## 3D Histograms

```
MatND histogram;
int channel_numbers[] = { 0, 1, 2 };
int* number_bins = new int[image.channels()];
for (ch=0; ch < image.channels(); ch++)
    number_bins[ch] = 16;
float ch_range[] = { 0.0, 255.0 };
const float* channel_ranges[] = { ch_range, ch_range, ch_range };
calcHist( &image, 1, channel_numbers, Mat(), histogram,
         image.channels(), a_number_bins, channel_ranges );
```

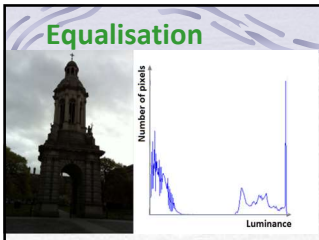
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## Equalisation

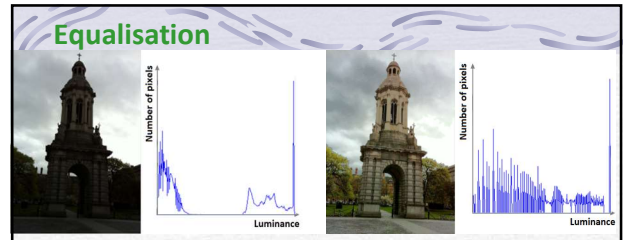


- If an image has insufficient contrast...
- Human can distinguish 700-900 greyscales
- How can we improve the contrast?
  - Evenly distribute values on all channels?
  - What happens to colours?

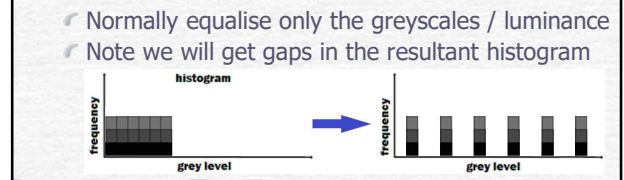
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## Equalisation



- Normally equalise only the greyscales / luminance
- Note we will get gaps in the resultant histogram



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## Equalisation

```
// Create a lookup table to map the luminances
// h[x] = histogram of luminance values image f(i,j).
pixels_so_far = 0
num_pixels = image.rows * image.cols

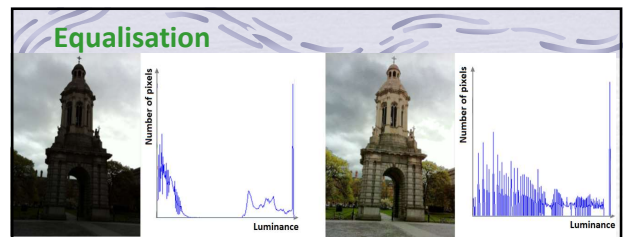
for input = 0 to 255
  pixels_so_far = pixels_so_far + h[ input ]
  LUT[ input ] = (pixels_so_far*256) / (num_pixels+1)

// Apply the lookup table LUT(x) to the image:
for every pixel f(i,j)
  g(i,j) = LUT[ f(i,j) ]
```

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## Equalisation



```
vector<Mat> channels( hls_image.channels() );
split(hls_image, channels);
equalizeHist( channels[1], channels[1] );
merge( channels, hls_image );
```

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## Histogram Comparison

- To find similar images...
  - Use metadata
  - Compare images?
- One simple possibility is to compare the colour distributions
  - Need a metric for comparisons...

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## Histogram Comparison



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## Histogram Comparison – Metrics

- $D_{Correlation}(h_1, h_2) = \frac{\sum_i (h_1(i) - \bar{h}_1)(h_2(i) - \bar{h}_2)}{\sqrt{\sum_i (h_1(i) - \bar{h}_1)^2 \sum_i (h_2(i) - \bar{h}_2)^2}}$
- $D_{Chi-Square}(h_1, h_2) = \sum_i \frac{(h_1(i) - h_2(i))^2}{(h_1(i) + h_2(i))}$
- $D_{Intersection}(h_1, h_2) = \sum_i \min(h_1(i), h_2(i))$
- $D_{Bhattacharyya}(h_1, h_2) = \sqrt{1 - \frac{1}{\sqrt{\bar{h}_1 \cdot \bar{h}_2 \cdot N^2}} \sum_i \sqrt{h_1(i) \cdot h_2(i)}}$
- where
  - o  $N$  is the number of bins in the histograms,
  - o  $\bar{h}_k = \sum_i (h_k(i)) / N$

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## Histogram Comparison – Earth Mover

An alternative to the metrics is to compute the Earth Mover's Distance...

- Minimum cost for turning a distribution into another distribution

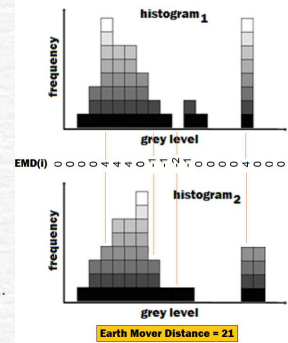
1D solution:

$$\text{EMD}(-1) = 0$$

$$\text{EMD}(i) = \text{histogram}_1(i) + \text{EMD}(i-1) - \text{histogram}_2(i)$$

$$\text{Earth Mover's Distance} = \sum_i |\text{EMD}(i)|$$

Colour EMD is harder to compute...



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## Histogram Comparison

```
normalize( histogram1, histogram1, 1.0);
normalize( histogram2, histogram2, 1.0);
double matching_score = compareHist( histogram1,
                                     histogram2, CV_COMP_CORREL);
```

We can also use Chi-Square (CV\_COMP\_CHISQR), Intersection (CV\_COMP\_INTERSECT) or Bhattacharyya (CV\_COMP\_BHATTACHARYYA) metrics or alternatively can use the Earth Mover's Distance function (EMD())

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## Histogram Back Projection

An approach to selecting colours (based on samples):

1. Obtain a representative sample set of the colours.
2. Histogram those samples.
3. Normalize that histogram so that the maximum value is 1.0.
4. Back project the normalized histogram onto any image  $f(i,j)$ .
5. This will provide a 'probability' image  $p(i,j)$  which indicates the similarity between  $f(i,j)$  and the sample set.



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## Histogram Back Projection in OpenCV



```
calcHist( &hls_samples_image, 1, channel_numbers, Mat(),
          histogram, image.channels(), number_bins, channel_ranges);
normalize( histogram, histogram, 1.0);
Mat probabilities = histogram.BackProject( hls_image );
```

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